

Abstract

Adder architectures play an important role as the basis for arithmetic circuits. This work proposes two new hybrid Full-Adder -FA- and a new hybrid Half-Adder -HA- circuits based on a recent XOR gate and a proposed AND gate. Several adder architectures presented in previous work are compared with the ones proposed in terms of delay, power consumption, number of transistors and number of bits. Extensive HSPICE simulations were performed for 65nm, 32nm, and 22nm. Based on the simulations it was concluded that the proposed FA elaborated using a recent XOR gate and the proposed AND gate presented a poor performance as technology scaled down. On the other hand, the proposed HA using recent XOR gate and proposed AND gate had superior speed and power consumption in the same simulations as the technology scaled down, presenting itself as an interesting candidate for future technologies.

Introduction

In the last decades, higher performance electronics have become part of our everyday life due to the development of MOSFETs and its continuous scaling down in size [1]. With the increasing demand for faster and low-energy-consuming electronic devices, designers attempt diverse approaches to obtain those desired circuit features [2].

Given that every mathematical operation in a binary system requires the computation of sums, arithmetic circuits become a focal point in circuit designs. Many adder architectures are being developed, as a consequence, efficiency and performance improvements in this area are obtained.

In this work, a recent XOR/XNOR gate [14] is modified to generate an XOR-AND circuit. Using XOR-AND circuit we propose two new hybrid Full-Adder -FA- and a new hybrid Half-Adder -HA- circuit.

17 different Carry Look Ahead -CLA- architectures are compared, each one built with a modified FA circuit. 9 FA designs with different styles are used [4]-[12] in addition to six very recent XOR/XNOR FA designs [14] and the two proposed hybrid FA. These 17 architectures are implemented from 2-bits up to 64-bits. Power-delay product -PDP- is used as a metric for comparison purposes due to previous work [14].

In addition to the previously mentioned implementations, Kogge-Stone adder -KSA- architectures up to 64-bits are implemented using the proposed hybrid HA and a conventional CMOS HA, making a total of 20 different architectures.

Objective

To evaluate the performance of new proposed adder architectures as to determine whether these can be used for higher performance electronics.

Methodology

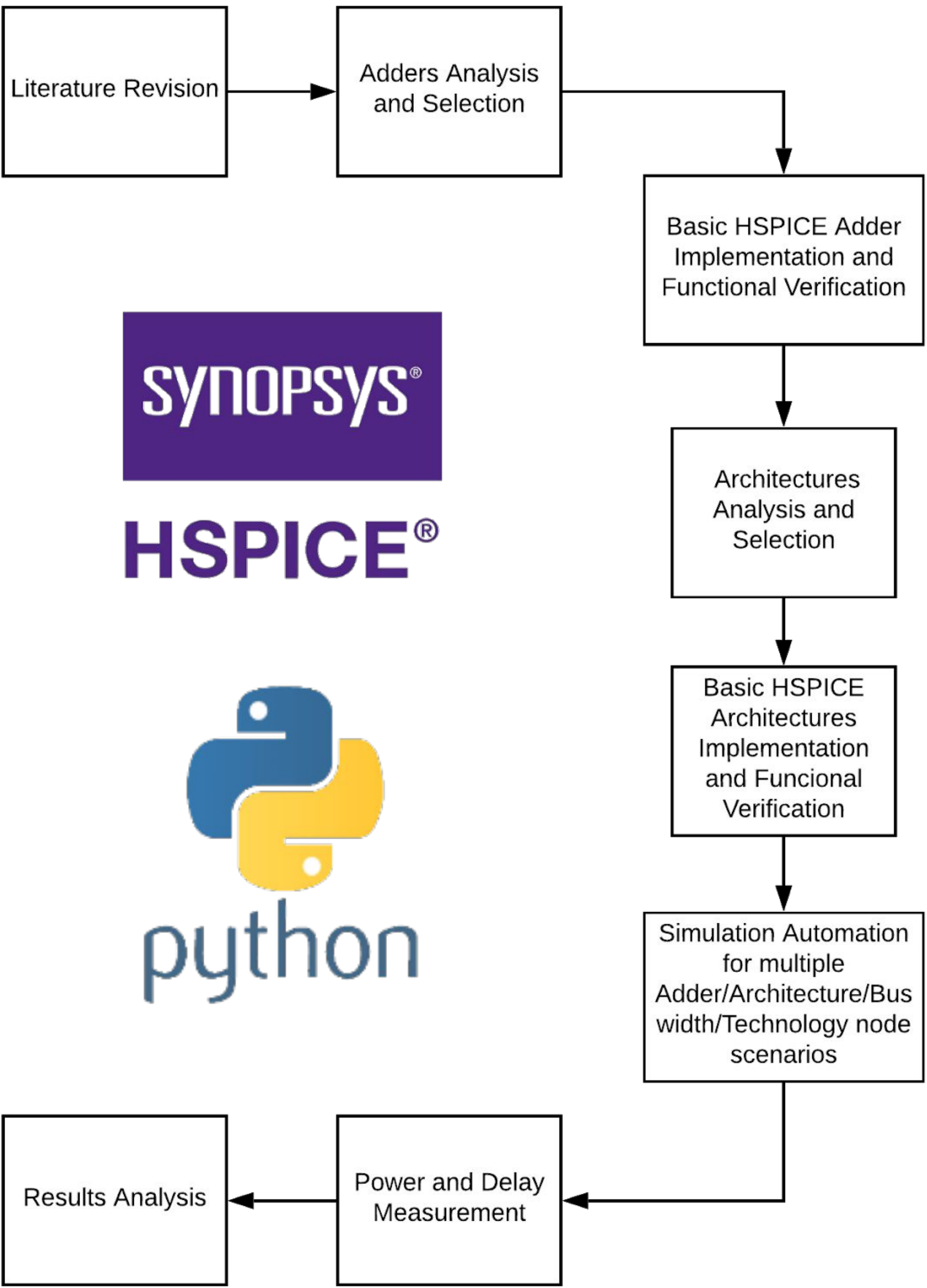


Fig. 1. Work Flow Diagram.

Proposed AND gate

By modifying the XNOR gate [14] we present a new AND gate, as shown in Figure 1. This new AND gate presents an advantage by using only four transistors for its implementation.

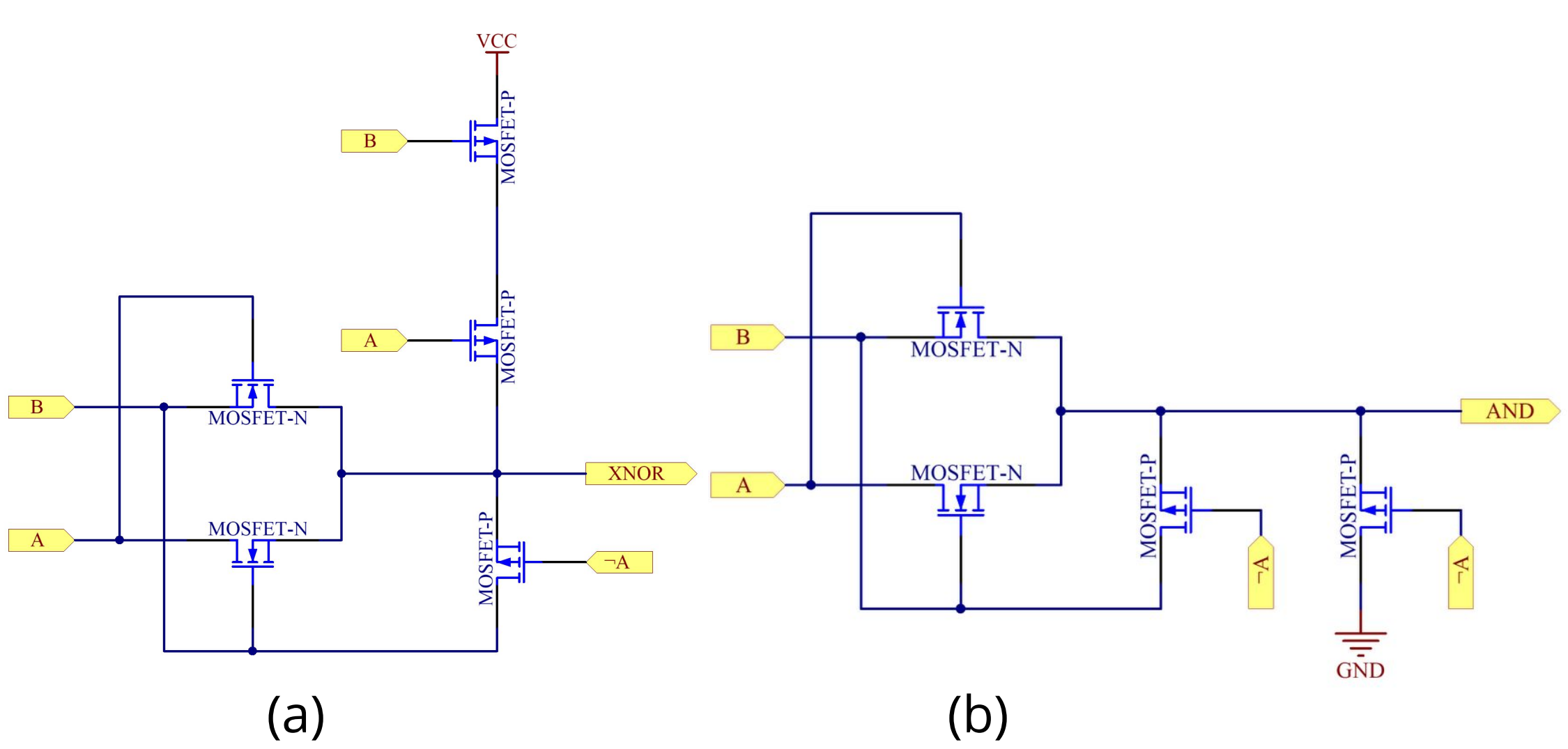


Fig. 2. (a) XNOR gate [14]. (b) Proposed AND gate.

Proposed FA and HA

To study the response of this new AND gate a hybrid FA was implemented by using standard CMOS OR gate, replacing the XOR gate with the one presented in [14] and replacing the AND gate with the one proposed in this work.

A new HA is also proposed by using the XOR gate [14] and the proposed AND gate. The schematics for the new proposed hybrid FA and HA are shown in Fig. 3(a) and Fig. 3(b) respectively.

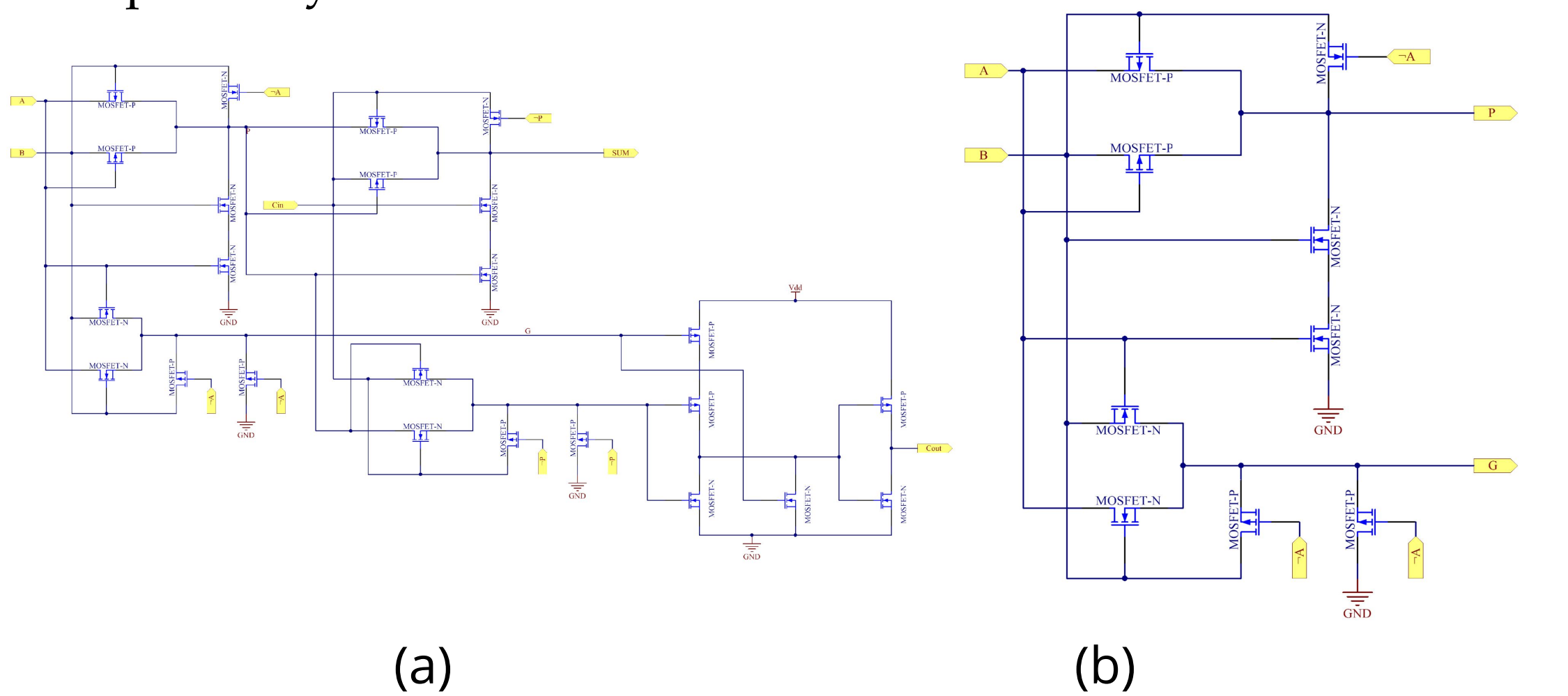


Fig. 3. (a) FA using new AND gate. (b) HA using new AND gate.

Another proposed structure is based on the use of standard CMOS FA using the XOR [14] while all other gates remain standard CMOS.

Adders implementation and functional verification

All FA and HA circuits were implemented on HSPICE for one-bit full simulations. CLA architectures were implemented using all 17 FA designs and KSA architectures were implemented for the proposed HA and the standard CMOS HA.

KSA presents a similar structure to CLA architecture, although the Fundamental Carry Operation -FCO- block rests among PG blocks and PPC blocks as shown in Fig. 4. Unlike the CLA, the KSA requires the previous Cout to compute the next value [13].

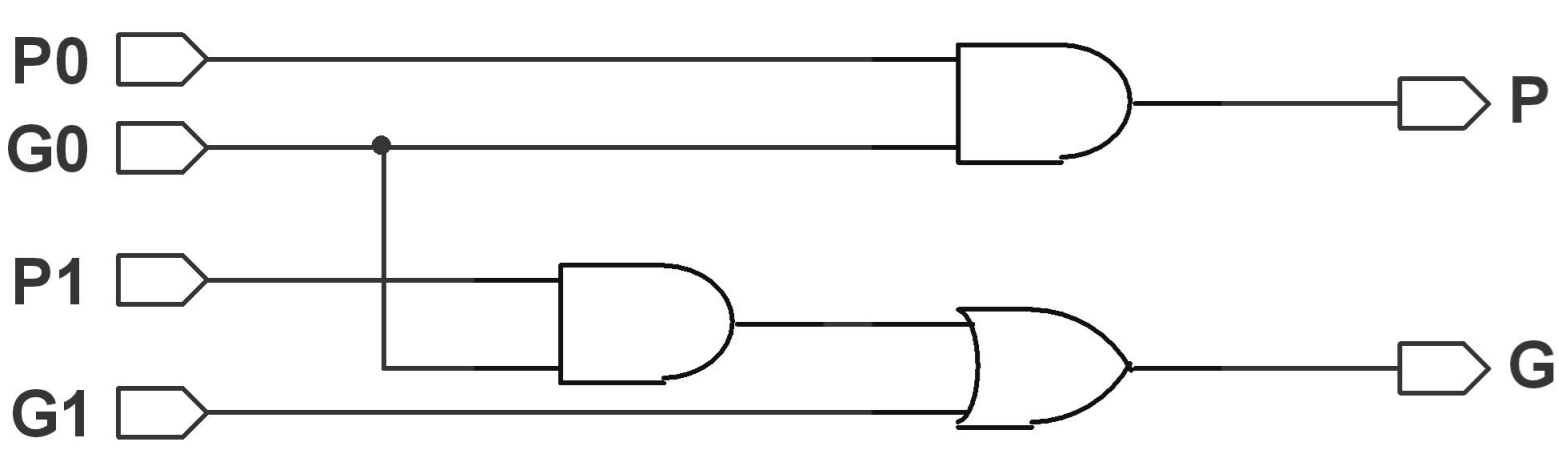


Fig. 4. FCO

Two KSA architectures are implemented using the proposed HA, differing in the AND gate for the FCO, one using the standard CMOS gate and the other using the proposed one.

CLA and KSA architectures from 2-bits up to 64-bits were implemented using 17 FA and 2 HA respectively. Each architecture was implemented using 3 different technologies, giving an approximate of 360 independent circuits for 120 architectures/microarchitectures.

Different algorithms capable of performing HSPICE simulations were elaborated to automate the generation of results for over a thousand simulations.

Results

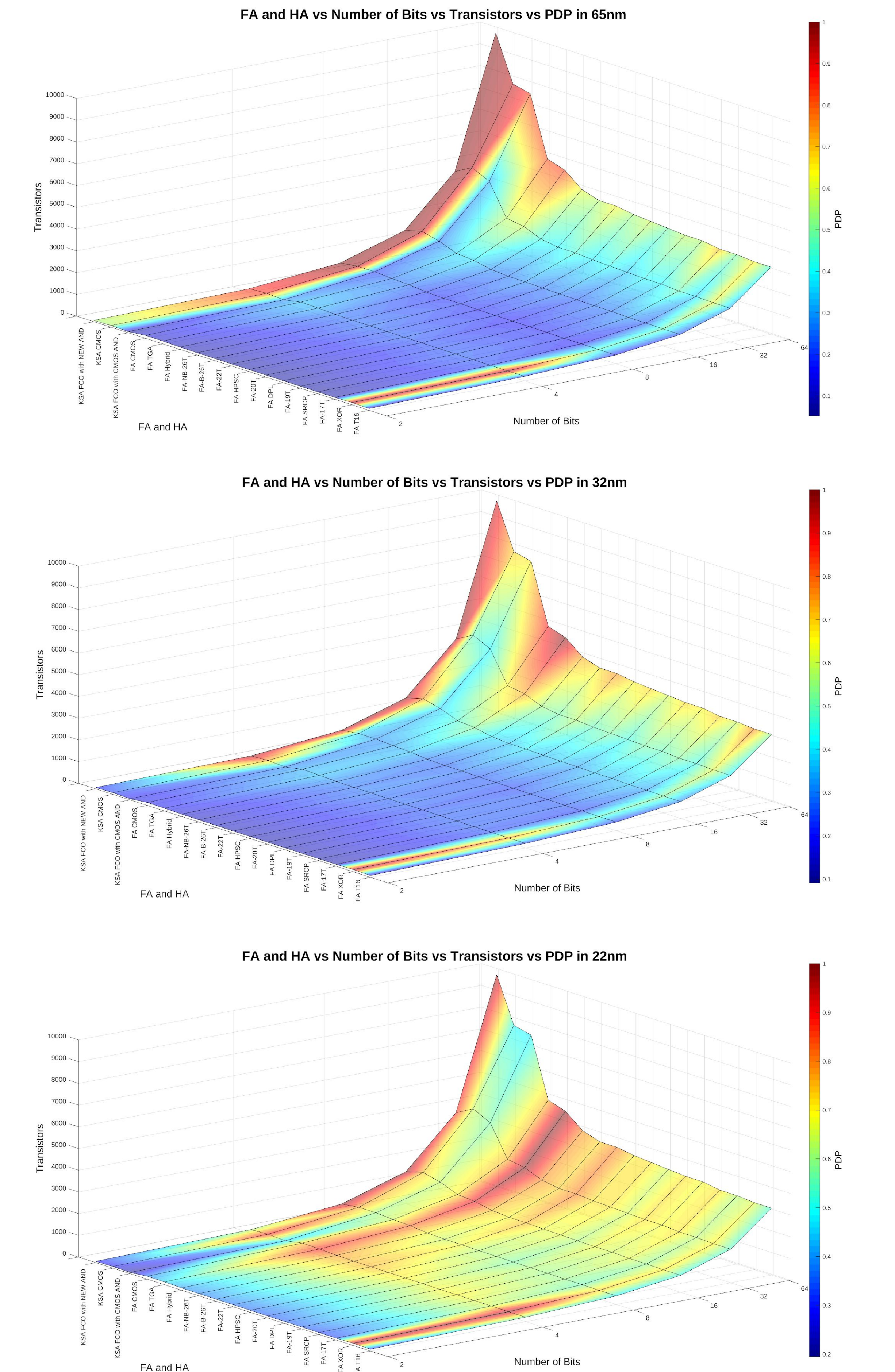
Heat Surfaces show the FA’s and HA’s behavior as technology scales down and the number of bits increased for all 20 architectures. PDP is presented as the heat variable. Some architectures were removed due to poor performance causing distortion.

As technology scaled down lowered performance -PDP- is shown for most adders while there are some exceptions where it improved.

KSA architectures required the largest amount of transistors for their implementation, while the proposed hybrid FAs required the least.

T16 FA CLA presents the lowest PDP for 65nm and 32nm. As technology scaled down, proposed hybrid KSA using proposed HA and standard CMOS AND gate in FCO presents the lowest PDP among all 22nm simulations.

CLA configuration using TFA FA in 65 and 32nm presented the largest PDP, as technology scales down CPL and the proposed hybrid FA using XOR [14] and proposed AND present a PDP increase, thus the proposed hybrid FA presents the largest PDP for 22nm technology.



Conclusions

Using the XOR [14] and the proposed AND gate to elaborate a hybrid FA resulted in an increase in its power consumption as technology scaled down. A larger power consumption was also presented by KSA using hybrid HA and the proposed AND gate in FCO, based on these results it is not recommended to use the proposed AND gate in a cascade style since it has a considerable negative impact on its performance.

The second FA implemented using the XOR [14] and standard CMOS AND gate showed a PDP decrease as the number of bits increased, being this the only FA presenting this type of behavior, which makes it a very interesting architecture for current or future wide-bus implementations.

CLAs implemented using FAs presented in [14] show better PDP when compared to all CLAs implemented in this work.

The hybrid KSA using proposed HA and standard CMOS AND gate in FCO presented a very large PDP at 65nm but consistently decreases as technology scaled down, becoming the one with the lowest PDP at 22nm. This behavior makes it interesting for possible future technologies.

Transistor sizing technique should be applied to all architectures in order to study possible further improvement.

References

Paper

